

Molecular Docking Analysis of Top 10 Hubs and Naringenin

Table 1. Structural information of naringenin and the selected hub proteins used for molecular docking analysis.

Ligand	SMILES	Protein Target	PDB ID	Resolution
Naringenin (CID: 439246)	<chem>C1[C@H](OC2=CC(=CC(=C2C1=O)O)O)C3=CC=C(C=C3)O</chem>	SRC	1Y57	1.91 Å
		AKT1	4EKL	2.00 Å
		PTGS2	5IKQ	2.41 Å
		MMP9	5I12	1.59 Å
		ESR1	1GWQ	2.45 Å
		BCL2	6O0K	1.62 Å
		PIK3CA	7R9V	2.69 Å
		CYP19A1	5JKV	2.75 Å
		MMP2	1GEN	2.15 Å
		APP	1AAP	1.50 Å

Molecular docking is a computational approach often used in drug discovery to help in predicting the structural details of proteins and protein-ligand interaction (Meng *et al.* 2011). To validate the interactions predicted by network pharmacology, molecular docking was performed between naringenin and the 10 hub proteins identified from the PPI network. Protein structures were obtained from Protein Data Bank (PDB), selecting the X-Ray crystallographic method with a resolution of ≤ 2.00 Å if available, or the highest-quality structure with the lowest available resolution was chosen (Table 1). The ligand-binding pockets of each protein target were predicted using PrankWeb. Prior to docking, protein structures were prepared by removing heteroatoms that might disturb the binding during docking simulation, whereas essential cofactors and metal ions important in structural stability and catalytic activity were retained. Docking simulations were conducted using SwissDock with the AutoDock Vina scoring function.

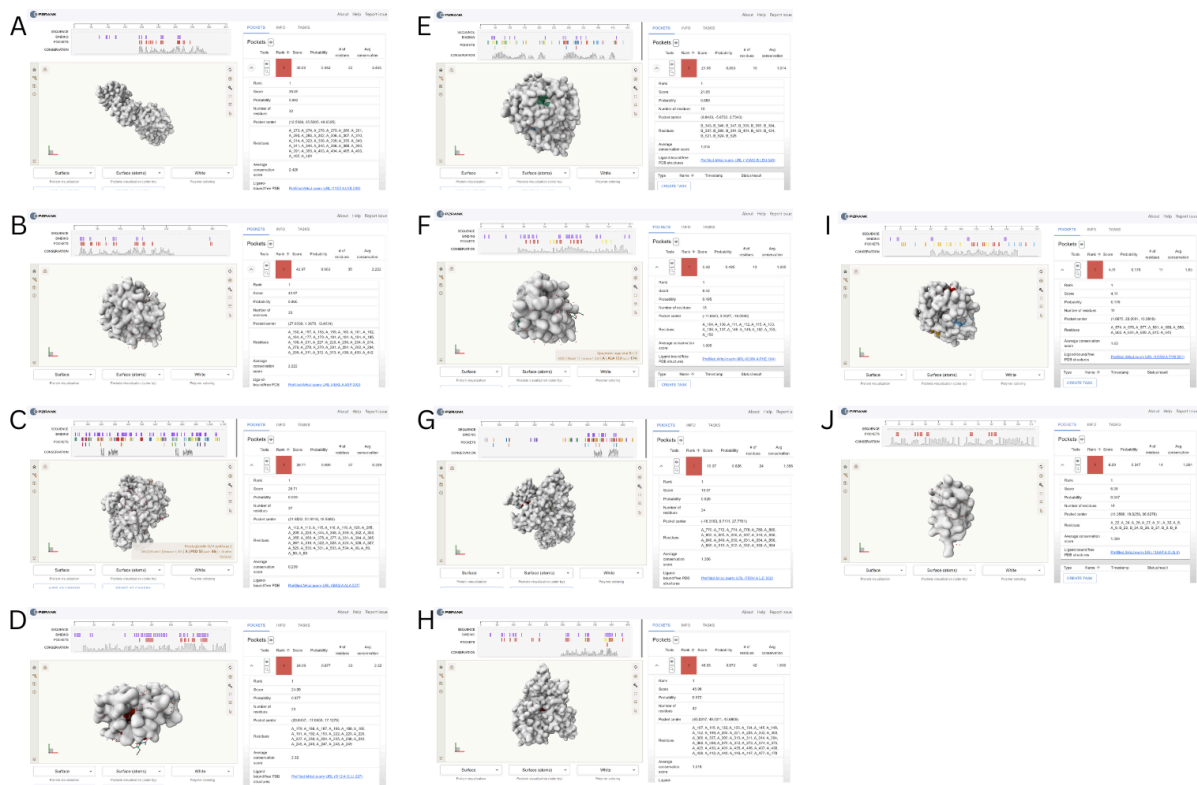


Figure 1. Ligand-binding pocket prediction using PrankWeb of SRC (A), AKT1 (B), PTGS2 (C), MMP9 (D), ESR1 (E), BCL2 (F), PIK3CA (G), CYP19A1 (H), MMP2 (I), and APP (J).

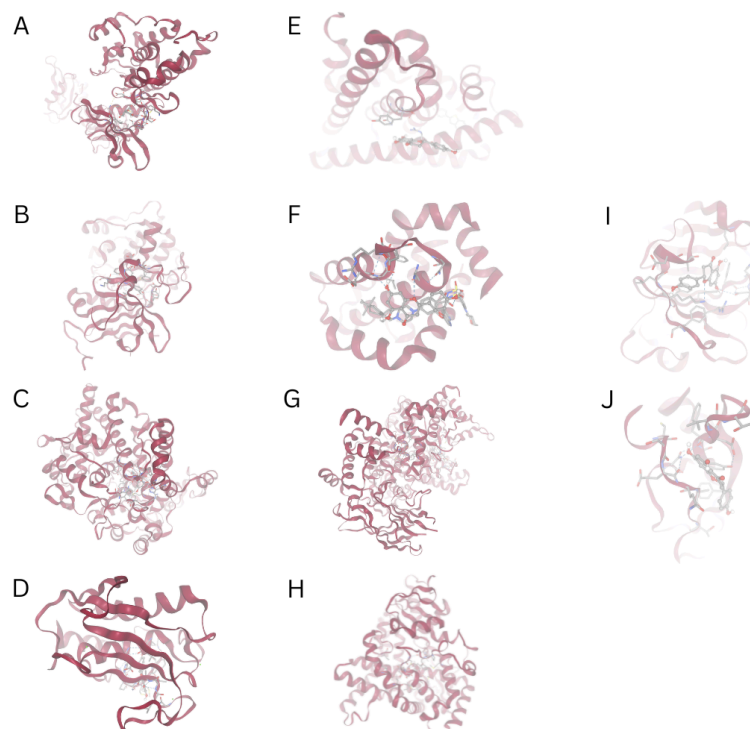


Figure 2. Docking Visualization using SwissDock of SRC (A), AKT1 (B), PTGS2 (C), MMP9 (D), ESR1 (E), BCL2 (F), PIK3CA (G), CYP19A1 (H), MMP2 (I), and APP (J).

A	Model	Calculated affinity (kcal/mol)
	1	-8.029
	2	-7.875
	3	-7.628
	4	-7.385
	5	-7.468
	6	-7.485
	7	-7.323
	8	-7.159
	9	-6.772
	10	-6.742
	11	-6.728
	12	-6.661
	13	-6.581
	14	-6.537
	15	-6.509
	16	-6.425
	17	-6.420
	18	-6.235
	19	-6.189
	20	-6.964

C	Model	Calculated affinity (kcal/mol)
	1	-8.081
	2	-7.744
	3	-7.694
	4	-7.673
	5	-7.454
	6	-7.376
	7	-7.326
	8	-7.206
	9	-7.100
	10	-6.989
	11	-6.589
	12	-6.199
	13	-5.911
	14	-5.348
	15	-5.214

E	Model	Calculated affinity (kcal/mol)
	1	-1.854
	2	-1.815
	3	-1.680
	4	-1.596
	5	-1.546
	6	-1.542
	7	-1.504
	8	-1.491
	9	-1.387
	10	-1.385
	11	-1.346
	12	-1.309
	13	-1.300
	14	-1.274
	15	-1.227
	16	-1.214
	17	-1.205
	18	-1.162
	19	-1.055
	20	-1.041

G	Model	Calculated affinity (kcal/mol)
	1	-6.362
	2	-6.303
	3	-6.120
	4	-5.907
	5	-5.873
	6	-5.826
	7	-5.674
	8	-5.562
	9	-5.518
	10	-5.455
	11	-5.418
	12	-5.401
	13	-5.370
	14	-5.195
	15	-4.713
	16	-6.975
	17	-6.945
	18	-6.913
	19	-6.907
	20	-6.377

I	Model	Calculated affinity (kcal/mol)
	1	-5.523
	2	-5.355
	3	-5.283
	4	-5.210
	5	-5.103
	6	-5.084
	7	-5.053
	8	-4.954
	9	-4.844
	10	-4.811
	11	-4.839
	12	-4.827
	13	-4.778
	14	-4.753
	15	-4.745
	16	-4.722
	17	-4.719
	18	-4.697
	19	-4.670
	20	-4.637

B	Model	Calculated affinity (kcal/mol)
	1	-8.773
	2	-7.821
	3	-7.787
	4	-7.684
	5	-7.538
	6	-7.418
	7	-7.418
	8	-7.337
	9	-7.315
	10	-7.311
	11	-7.287
	12	-7.267
	13	-7.054
	14	-7.027
	15	-6.934
	16	-6.886
	17	-6.882
	18	-6.819
	19	-6.734

D	Model	Calculated affinity (kcal/mol)
	1	-9.239
	2	-8.881
	3	-8.409
	4	-7.969
	5	-7.737
	6	-6.932
	7	-6.832
	8	-6.728
	9	-6.428
	10	-6.370

F	Model	Calculated affinity (kcal/mol)
	1	-6.183
	2	-6.098
	3	-5.921
	4	-5.847
	5	-5.694
	6	-5.387
	7	-5.385
	8	-5.359
	9	-5.329
	10	-5.288
	11	-5.328
	12	-5.094
	13	-4.977
	14	-4.877
	15	-4.807
	16	-4.753
	17	-4.689
	18	-4.657
	19	-4.609

H	Model	Calculated affinity (kcal/mol)
	1	-7.657
	2	-7.489
	3	-7.480
	4	-7.225
	5	-7.220
	6	-7.219
	7	-7.212
	8	-7.196
	9	-7.094
	10	-7.087
	11	-7.080
	12	-6.987
	13	-6.906
	14	-6.887
	15	-6.885
	16	-6.876
	17	-6.864
	18	-6.820
	19	-6.795

J	Model	Calculated affinity (kcal/mol)
	1	-4.901
	2	-5.303
	3	-5.288
	4	-5.213
	5	-5.189
	6	-5.153
	7	-5.150
	8	-4.988
	9	-4.984
	10	-4.963
	11	-4.772
	12	-4.682
	13	-4.676
	14	-4.651
	15	-4.611
	16	-4.610
	17	-4.573
	18	-4.548
	19	-4.501

Figure 3. Binding affinity calculation using SwissDock of SRC (A), AKT1 (B), PTGS2 (C), MMP9 (D), ESR1 (E), BCL2 (F), PIK3CA (G), CYP19A1 (H), MMP2 (I), and APP (J).

The predicted binding pockets (Figure 1), were generally located within the cavities of the proteins. These pockets resemble the ligand-binding regions and served as the search space for molecular docking. The visualization of the docking results (Figure 2) displays that naringenin successfully binds the predicted active sites in each protein target. This suggests that the shape and size of the binding sites are compatible with naringenin. Furthermore, the calculated affinity (Figure 3) shows a notable variation of affinities among the protein targets. The strongest interaction was observed with MMP9, exhibiting the lowest binding energy of -9.239 kcal/mol, meaning the strongest binding. This was followed by AKT1 (-8.773 kcal/mol), PIK3CA (-8.383 kcal/mol), PTGS2 (-8.081 kcal/mol), SRC (-8.029 kcal/mol) and CYP19A1 (-7.657 kcal/mol). The binding affinities below -7.0 kcal/mol suggest relatively strong and stable protein-ligand interactions (Venkatesan & Muniyan 2025). Therefore, this suggests that naringenin is less likely to bind efficiently to ESR1, BCL2, MMP2, and APP, indicating that the interaction may not be sufficient to significantly alter protein function.

The docking results support the multi-target characteristics predicted by the network pharmacology analysis. Particularly, the strong binding affinities with MMP9, AKT1, PIK3CA, PTGS2, SRC, and CYP19A1. These proteins are known to regulate key processes involved in cervical cancer progression. MMP9 contributes to remodeling of the ectocervix (Ghosh *et al.* 2014). AKT1 protein is involved in cell proliferation, differentiation and apoptosis (Qiu *et al.* 2025). PIK3CA mutations in cervical cancer are associated with higher tumor mutation burden (Voutsadakis 2021). PTGS2 protein might be affecting macrophage polarization and mutation effects (Zou *et al.* 2022). SRC protein is one of the most studied oncoproteins, known to be the one that can orchestrate the metabolism pathway to support specific tumour cell processes (Pelaz & Tabernero 2022). CYP19A1 is upregulated in cervical cancer to promote local estrogen production, driving tumor cell proliferation and development (Setiawan *et al.* 2009). In conclusion, this study provides structural evidence that suggests naringenin may exert its anticancer activity through simultaneous modulation of the protein targets that hold a key role in cervical cancer progression.

Reference

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